Physics-Informed Neural Networks and Neural-Mechanistic Hybrids for a WNT–RA–HOX Model of Colorectal Cancer Stemness

From a nondimensionalized mechanistic 7-ODE model to sparse-data forward solves, Bayesian identifiability, and learned terms.

Amirkhan Aidarkhan · Pascal Kataboh

01 The model

Seven coupled ODEs for the WNT–RA–HOX axis, nondimensionalized to **36 parameters** across four disease regimes.

02 The PINN

A Fourier-feature network solves the system forward from **40 observations**, then inverts it for the 36 parameters.

03 Bayesian inference

HMC turns the recovery ceiling into a diagnosis: which parameters the data can **never** pin down, and why.

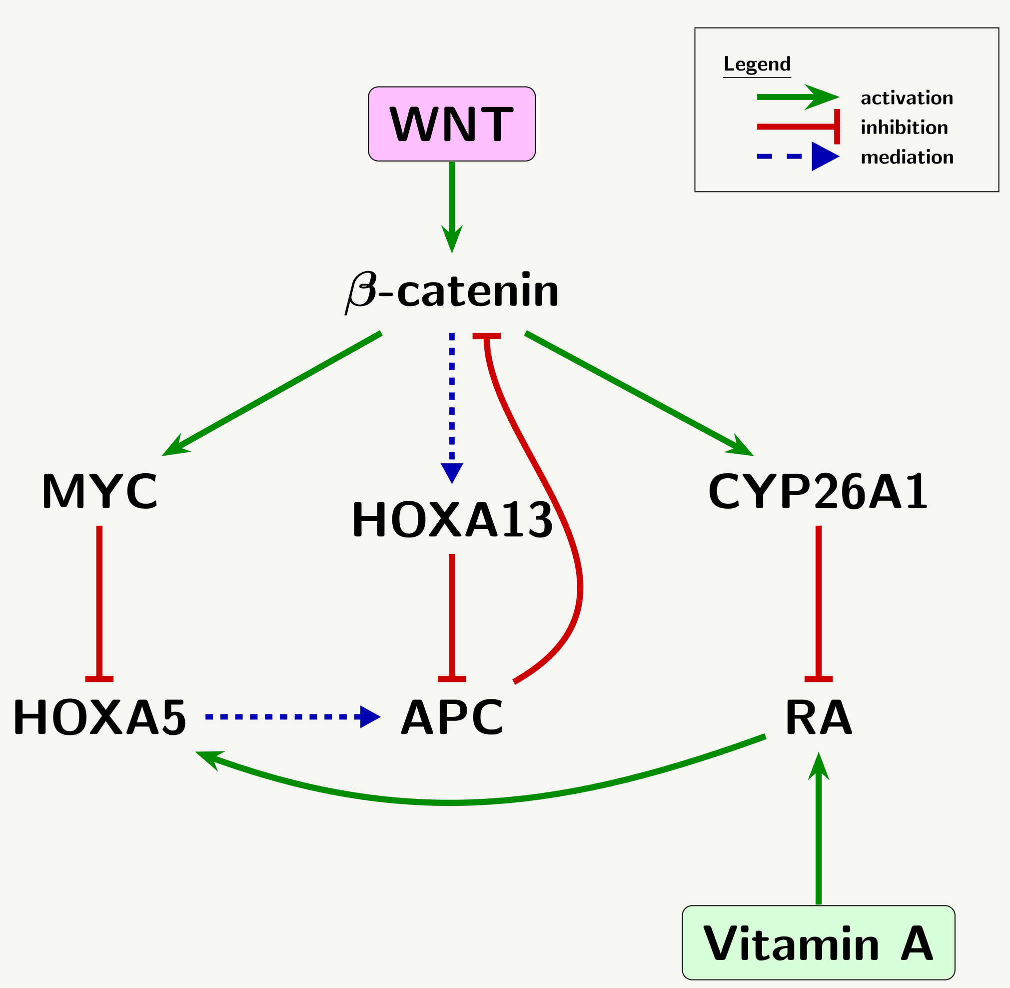
04 Neural-mechanistic hybrid

Replace one mechanistic term with a neural network, and the identifiability problem returns in a new form.

1 THE MODEL

Seven equations, nondimensionalized to 36 parameters.

In the intestinal crypt, WNT signalling drives proliferation, retinoic acid drives differentiation, and the HOX genes arbitrate between them. Loss of APC, the canonical adenoma-to-carcinoma driver, breaks that balance.



Green activates, red inhibits, dashed mediates. Seven species, one constitutive WNT drive, and a vitamin-A input.

Nondimensionalization

01 Scale each species by its characteristic value

taken to be its initial condition, and scale time by the reference β-catenin degradation rate $d_B = 1\ \mathrm{hr}^{-1}$:

$$B = B_0 b,\quad P = P_0 p,\quad \ldots,\quad \tau = d_B t$$

β-catenin is WNT's direct target and drives most other species, so its turnover sets the network's reference timescale.

02 Every equation collapses to one form

each loss term normalizes to $-x$, and the prefactor is a **pure timescale** ratio:

$$\varepsilon_X \frac{dx}{d\tau} = \text{production} - x$$

03 The parameter count falls to 36

Nondimensionalization changed the units and the count, not the model. The system is **stiff**: $\varepsilon_R = 0.40$, $\varepsilon_M = 0.60$.

The scientific readout: the stemness index

$$S(\tau) = \frac{b(1+\alpha_{13}h_{13})}{(1+p)(1+\alpha_5 h_5)}$$

Large when proliferative species dominate, small when differentiating species do. Not a state variable, but the aggregate the model exists to predict.

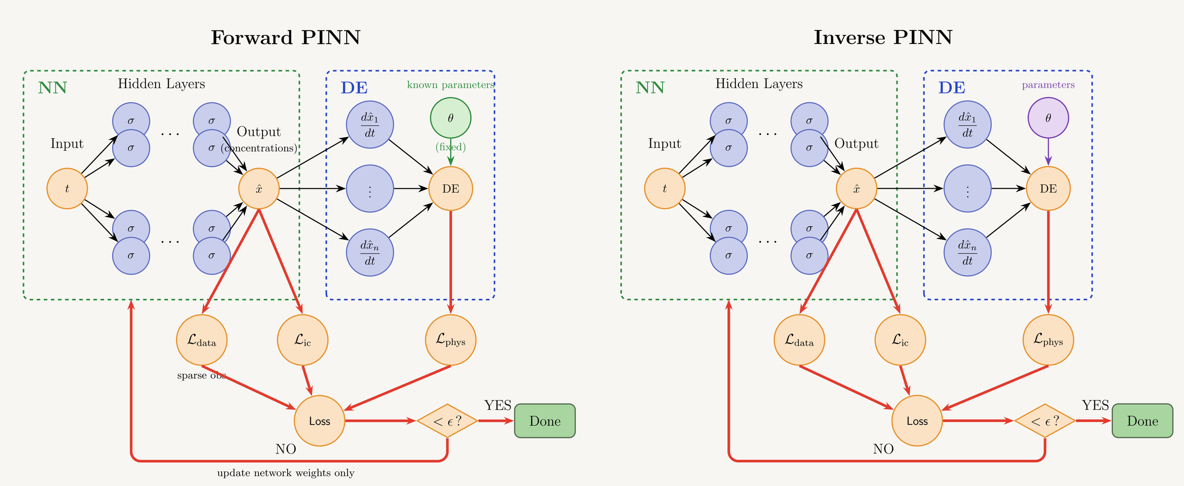
Four regimes, two parameters

Normal → Early Adenoma → Advanced Adenoma → Severe APC Loss differ **only** in the WNT drive W (0.8 → 2.0) and APC functionality θ_P (1.0 → 0.25).

2 THE PINN

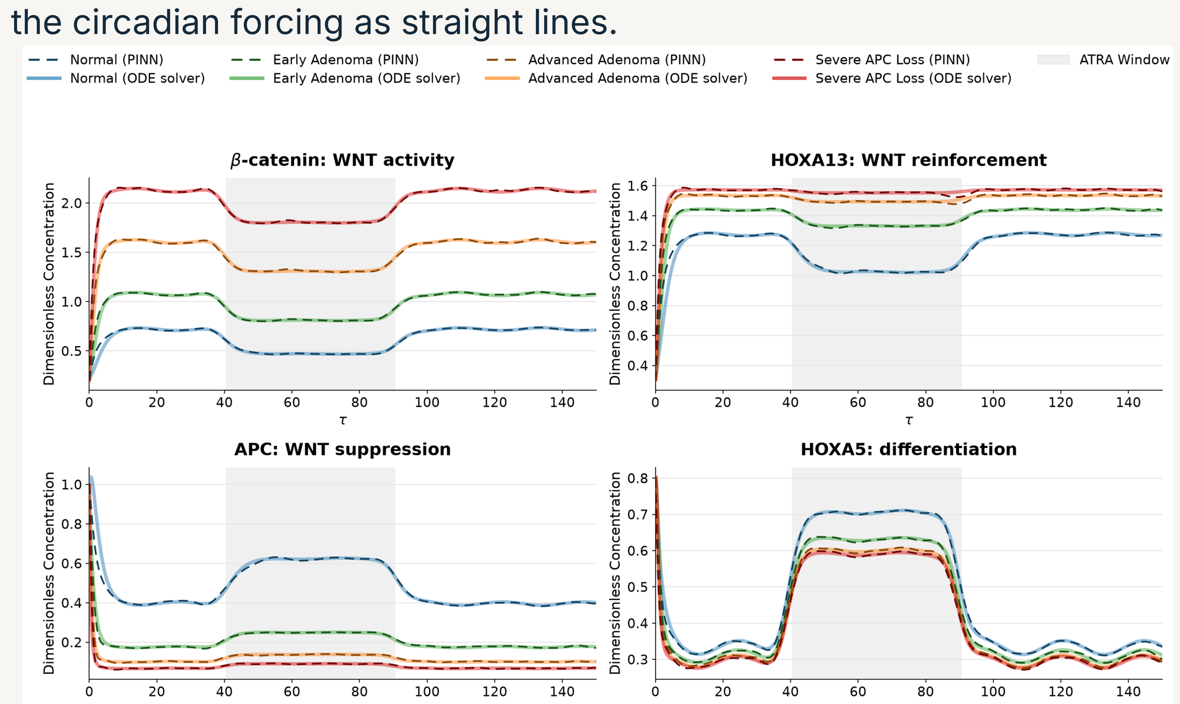
One network solves it from 40 points, and inverts it.

FORWARD · θ fixed   INVERSE · θ trainable



The network proposes a trajectory; automatic differentiation feeds it back through the ODEs. Three losses: data, initial condition, physics. Freeing θ turns the solver into a parameter estimator.

A Fourier-feature time embedding (16 modes, σ = 4) into a 256-wide, 4-deep GELU network, 207,879 weights. Without it, spectral bias renders the circadian forcing as straight lines.

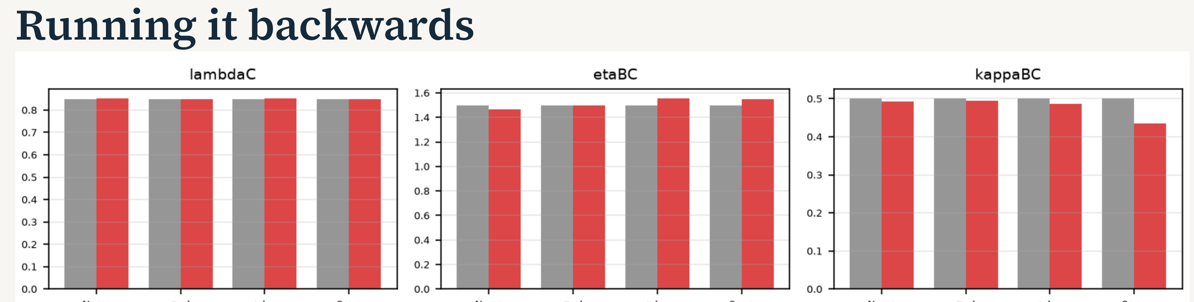


Trained on **40 sparse observations** plus the physics residual, the PINN tracks the Radau reference through the ATRA window in every regime.

1.06%   **2.41%**

dense-label baseline, grand mean rel-L2   from 40 observations, the honest solve

Running it backwards



True (grey) against recovered (red), three of the best-converging parameters. Recovery is ill-posed at 36: the PINN-specific ceiling is 8/36, from gradient starvation and **autodiff derivative bias**.

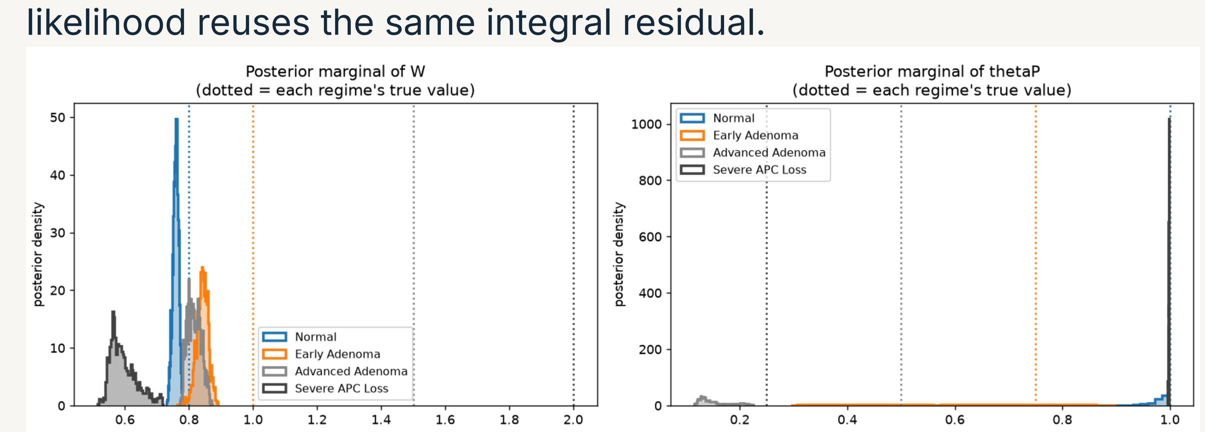
A derivative-free integral residual breaks half that ceiling

Enforcing the ODE in integrated form uses network **values** only, never the biased derivative. Recovery goes **10/4/5/7 → 17/16/10/7**, a same-conditions total of **37 → 50** of 144. Severe APC Loss does not move; what remains is structural, not numerical.

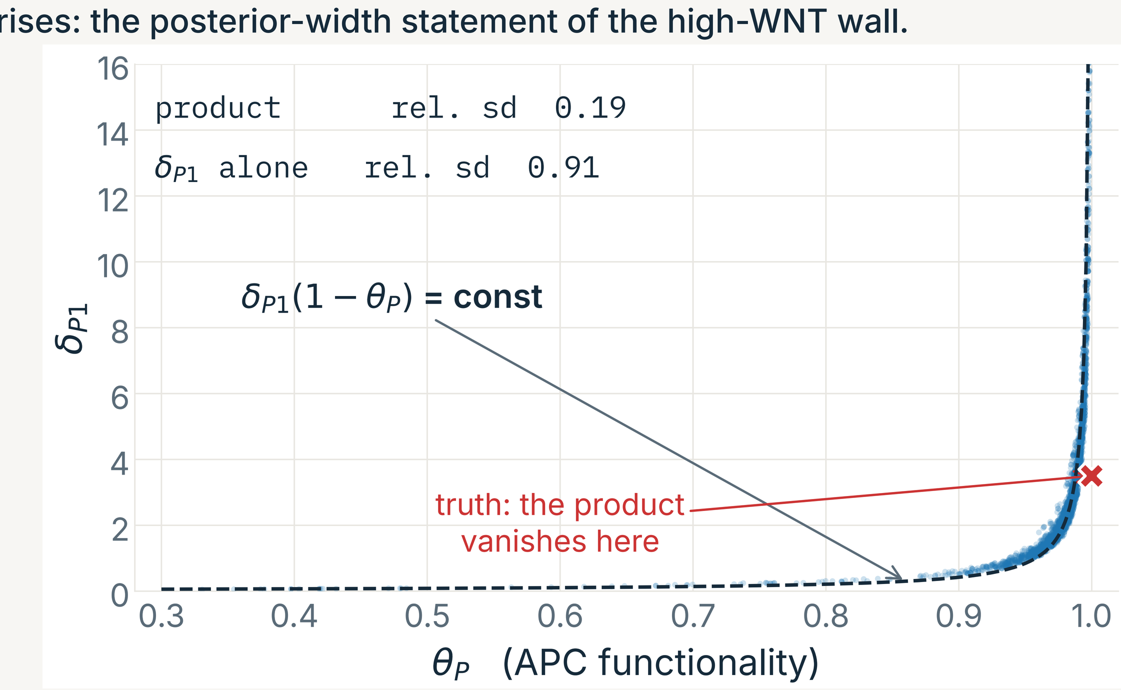
3 BAYESIAN INFERENCE

A posterior says what the data can never pin down.

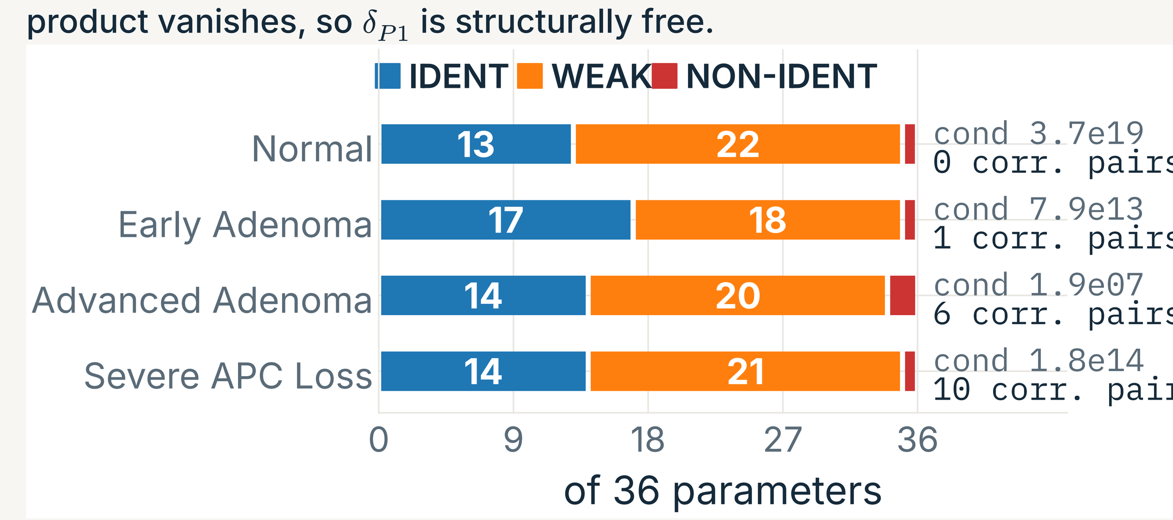
A point estimate cannot say **why** a parameter failed. We run HMC over the 36 parameters with the state networks frozen: the Bayesian twin of the point refine stage, and derivative-free, because the likelihood reuses the same integral residual.



W stays tight in every regime. θ_P reverts toward its prior as the WNT drive rises: the posterior-width statement of the high-WNT wall.



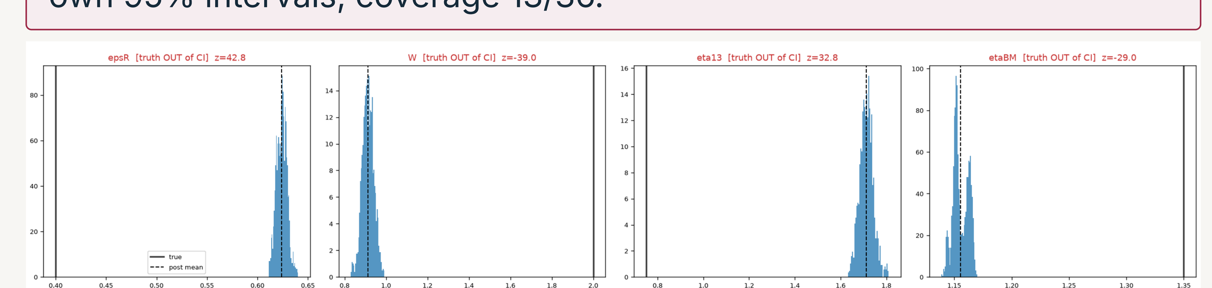
Unprompted, the sampler rides a curved valley: the **product** $\delta_{P1}(1-\theta_P)$ is 4.7× better constrained than δ_{P1} alone. At the healthy truth $\theta_P = 1$ the product vanishes, so δ_{P1} is structurally free.



The Fisher information agrees from a third direction: conditioning degrades with WNT drive, to 3.7 × 10¹⁹ in Severe APC Loss.

The first run said 36/36. It was an artifact.

The physics likelihood summed 210k residual terms against a tiny σ, producing a pinpoint, biased posterior. Below: truths far outside their own 95% intervals, coverage 13/36.



Honest recovery is **19/20/11/8**, not 36/36. The run still fails its own effective-sample-size and coverage gates, so we claim the **geometry** and the diagnosis, never the marginal widths.

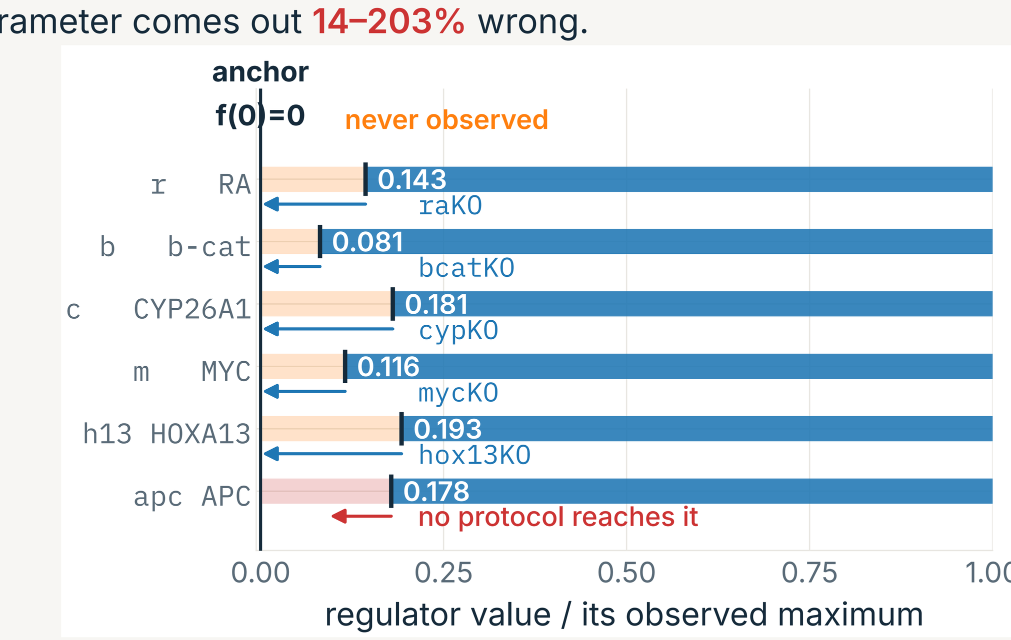
4 NEURAL-MECHANISTIC HYBRID

Replace a term with a network and identifiability returns.

$$\varepsilon\dot{x} = a + f_{\mathrm{NN}}(u) - x$$

the anchor $f_{\mathrm{NN}}(0) = 0$ is meant to stop the network absorbing a constant out of the basal-production parameter a

It does not work: across **five** parameterisations, including the monotone-and-bounded one the UDE literature recommends, the basal parameter comes out 14–203% wrong.



Because across all 10 conditions and all 4 regimes, **no regulator ever approaches zero**. The anchor is a constraint asserted at a point the data never visit.

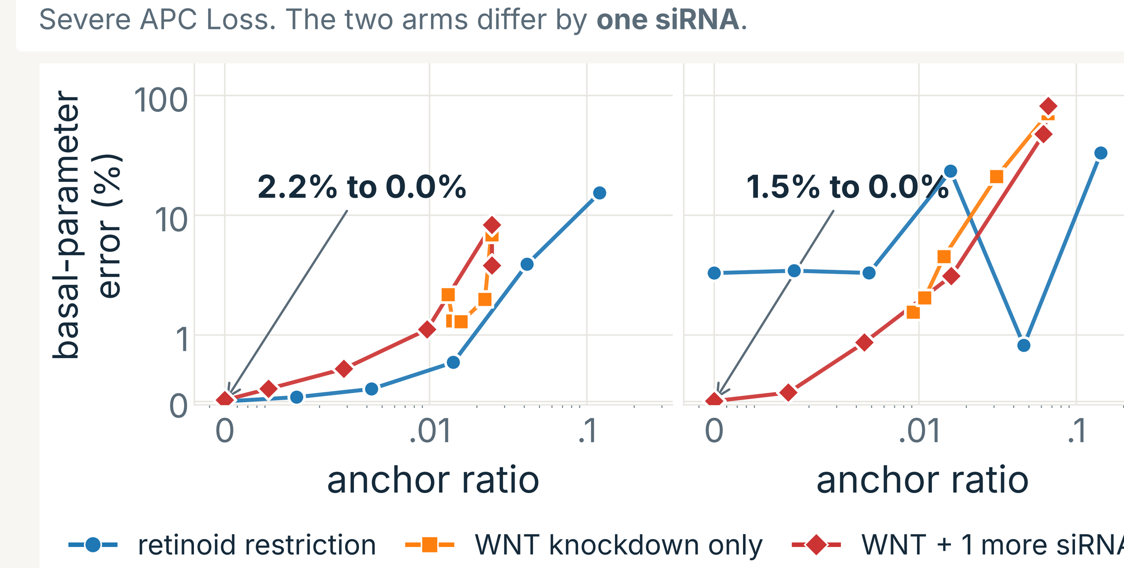
The fix is an experiment, not an architecture

A dose–response with the **condition count held fixed at eleven** at every dose, so "more data helps" is ruled out by construction.

misses the anchor   reaches it · one more siRNA

2.2% / 1.5%   **0.0% / 0.0%**

Same edge, same equation, same parameter, same seeds, same budget. Normal / Severe APC Loss. The two arms differ by **one siRNA**.



Against the **anchor ratio** the arms collapse onto one curve; against dose they do not. The anchor, not the intervention, is the governing variable.

Takeaways

1. Nondimensionalization plus Fourier features makes a stiff, multiscale 7-ODE system solvable by a PINN from sparse data.
2. Inverse recovery is limited by **information**, not architecture, and a posterior says exactly which directions are unidentifiable.
3. A constraint on a learned term carries information only where the data have support; the remedy is experiment design, computable in advance.

What these numbers are not

Hybrid numbers come from an equation-local screen fed exact states, an upper bound though both arms are fitted identically so the differences are fair. Restart noise ±0.5–0.9 pp; single seed, so no error bars. Not shown: an information-matched control (no gain in 8 of 8 cells) and a prospective test (95.6% → 0.0%). 4 of 6 pre-registered predictions failed and are reported as failures.

References · Raissi, Perdikaris & Karniadakis, J. Comput. Phys. 378:686 (2019); Karniadakis et al., Nat. Rev. Phys. 3:422 (2021); Tancik et al., NeurIPS (2020); Yang, Meng & Karniadakis, arXiv:2003.06097 (2020); Jung & Choi, arXiv:2210.11737 (2022); Raue et al., Bioinformatics 25:1923 (2009); Engl et al., Inverse Problems 25:123014 (2009); Rackauckas et al., arXiv:2001.04385 (2020); Philipps, Schmid & Hasenauer, npj Syst. Biol. Appl. 11:101 (2025); Loman & Baker, arXiv:2510.14140 (2025); Wang & Hill, IEEE Trans. Neural Netw. 17(1):130 (2006); Plate, Martensen & Sager, arXiv:2408.07143 (2024); Velioglu et al., Comput. Chem. Eng. (2025); Faure et al., Nat. Commun. 14:4213 (2023).